Is Batting Average Overrated?

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1 Introduction

Batting average is one of the most recognizable statistics in baseball. It is commonly displayed during broadcasts, alongside player names in scorebugs and lineup graphics, and has long been used as a measure of offensive performace. For many fans, batting average is often the first statistic considered when evaluating a hitter.

Despite its popularity, the value of batting average has become heavily debated in recent decades. Many analysts argue that metrics like on-base percentage (OBP) , slugging percentage (SLG), and on-base plus slugging (OPS) offer more accurate indicators of offensive contribution due to the growth of sabermetrics and advanced baseball analytics. Unlike batting average, these statistics account for factors such as walks and extra-base hits, which may have a greater impact on a team's ability to score runs and win games.

As a result, an important question emerges:

Is batting average still a meaningful indicator of team success, or have modern offensive metrics replaced its usefulness?

The goal of this project is to investigate the relationships between several offensive statistics and regular season wins in the MLB since 2000. Particular attention is given to batting average, on-base percentage, slugging percentage, OPS, walk rate, strikeout rate, and isolated power. By comparing the strength of the relationships between these statistics and team wins, this analysis seeks to whether batting average remains a valuable measure of team success compared to more modern metrics.

2 Data Description

The data used in this analysis were obtained from the Lahman Baseball Database, a publicly available collection of historical Major League Baseball statistics. The original dataset contains team-level records spanning more than a century of professional baseball history. To focus on the modern era of the sport, only observations from the 2000 season onward were retained. Additionally, the 2020 season was excluded from the analysis due to its shortened 60-game schedule resulting from the COVID-19 pandemic.

Each observation in the dataset represents a single Major League Baseball team during a particular season. After filtering, the dataset contained 750 team-seasons. Several offensive variables were selected for analysis, including at-bats (AB), hits (H), doubles (2B), triples (3B), home runs (HR), walks (BB), strikeouts (SO), sacrifice flies (SF), and regular season wins (W).

Using these variables, several commonly used offensive statistics were calculated. Batting average (AVG) was included as a traditional measure of hitting performance, while on-base percentage (OBP), slugging percentage (SLG), and on-base plus slugging (OPS) were included as more comprehensive measures of offensive production. In addition, walk rate, strikeout rate, and isolated power (ISO) were calculated to display different aspects of offensive approach and effectiveness.

Before doing any statistical analysis, the dataset was examined for missing values and inconsistencies. No missing observations were found among the variables used in the study.

3 Methods

To evaluate the relationships between offensive performance metrics and regular season success, a correlation-based approach was used.

First, several offensive statistics were calculated from the raw team batting data. This includes:

Batting Average: $avg = \frac{H}{AB}$

On-base Percentage: $obp = \frac{H+BB+HBP}{AB+BB+HBP+SF}$

Total Bases: $TB = H + 2B + 2 * (3B) + 3 * (HR)$

Slugging Percentage: $slg = \frac{TB}{AB}$

On-Base Plus Slugging: $ops = obp + slg$

Walk rate and Strikeout Rate: $\frac{BB}{PA}$ and $\frac{SO}{PA}$

Isolated Power: $iso = slg - avg$

These statistics were chosen because they represent different aspects of offensive performance, including a team's ability to generate hits, reach base, hit for power, draw walks, and avoid strikeouts.

Pearson correlation coefficients were then calculated between regular season wins and each offensive statistic. The Pearson correlation coefficient provides a measure of the strength and direction of a linear relationship between two variables, taking values between -1 and 1. Values close to 1 indicate a strong positive association, values close to -1 indicate a strong negative association, and values near 0 suggest little or no linear relationship.

After calculating the correlation coefficients, the offensive statistics exhibiting the strongest relationships with wins were identified. Particular attention was given to comparing batting average with more modern offensive metrics such as OBP, SLG, and OPS. This comparison was motivated by the ongoing debate within baseball analytics regarding the usefulness of batting average as a measure of offensive performance.

To complement the numerical analysis, scatterplots were created to visualize the relationships between selected offensive statistics and regular season wins. These visualizations were used to identify overall trends, compare the strength of relationships, and highlight notable team-seasons that appeared to deviate from the general patterns observed in the data.

4 Results

4.1 Batting Average

The first metric result I want to show is the main topic of the video: Batting Average. For those who don't know, batting average measures how many hits a player gets per at-bat.

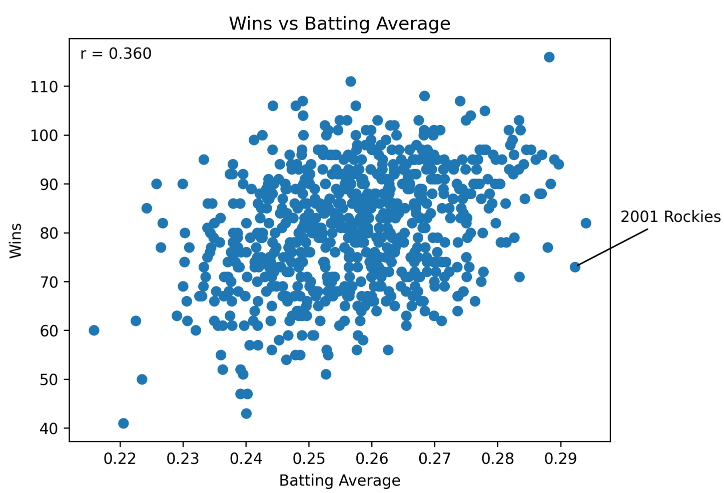

Figure 1: Relationship between batting average and regular season wins.

Observing the graph, it is quite clear that there is not a very strong linear relationship between batting average and wins. The relatively weak r-value of 0.360 suggests this, along with a large amount of teams that prove to be great examples of why batting average alone is not a very accurate measure of success in the MLB.

The 2001 Colorado Rockies had an extremely impressive batting average of 0.292, the second-highest batting average out of any team analyzed. However, as is obvious in the graph they did not have a very successful year whatsoever, with a record of 73-89. The high batting average was driven by Coors Field's hitter-friendly environment, but poor pitching led to an unsuccessful season.

This is a great example of how batting average doesn't necessarily correlate to wins as much as some believe it does.

4.2 Metric Comparison

Here's how every other metric analyzed compares to batting average in terms of correlation with wins. This chart makes it clear that the correlation between batting average and wins is not nearly as strong as other modern metrics.

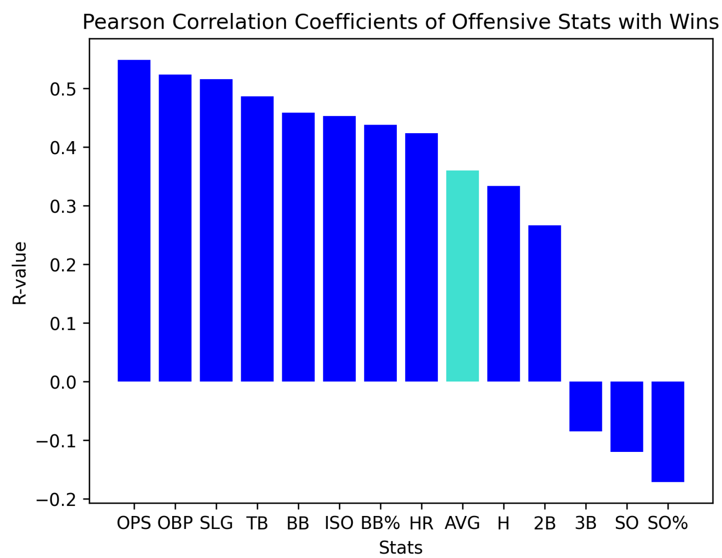

Figure 2: R-values for each offensive metric analyzed.

Let's look at the metrics that have the top 3 strongest correlations with wins, starting with Slugging Percentage in 3rd.

4.3 Slugging Percentage

Slugging Percentage is actually very similar to Batting Average, with one small difference. Slugging Percentage also measures how many hits a player gets per at bat, but weighs doubles, triples, and home runs increasingly heavier. This could be interpreted as a 'bases per at-bat' as opposed to hits per at bat.

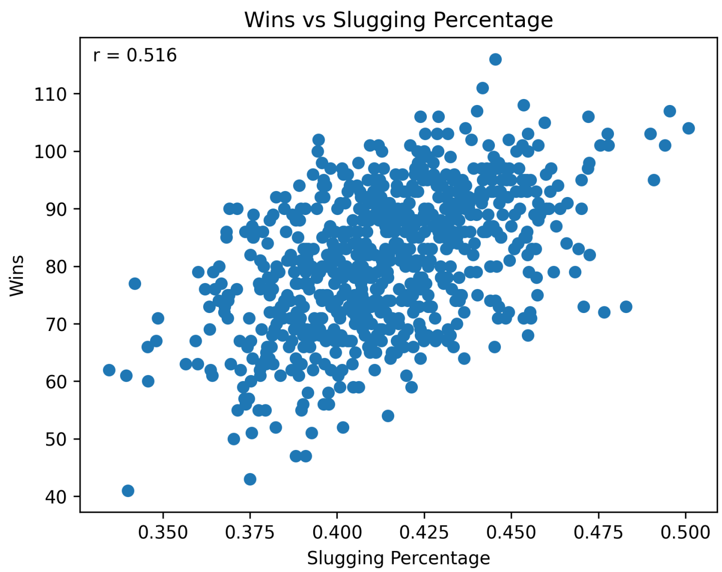

Figure 3: Relationship between slugging percentage and regular season wins.

The visual makes it very clear that slugging has a much stronger correlation with wins than batting average.

4.4 On-Base Percentage

In 2nd place, we have On-Base Percentage. This is a very simple metric, measuring how often a player gets on base per plate appearance, hence the name on-base percentage.

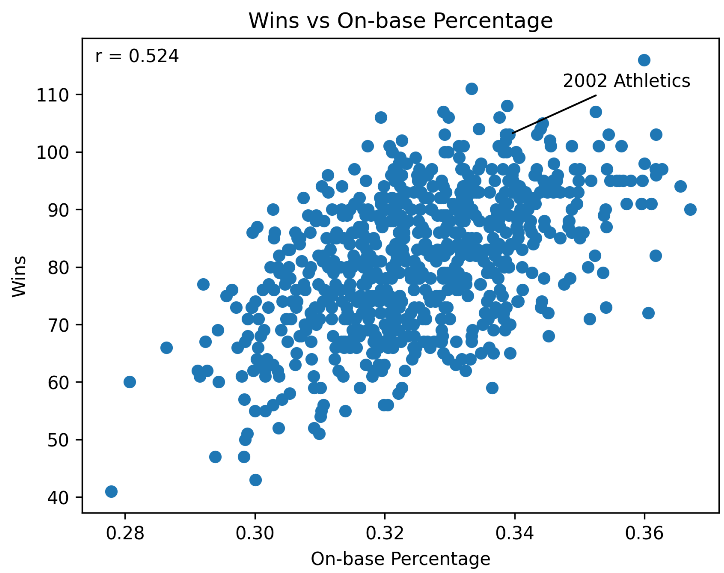

Figure 4: Relationship between on-base percentage and regular season wins.

The season pinned here is the 2002 Oakland Athletics, well-known for being the 'Moneyball' team who prioritized OBP rather than AVG and ended up winning 103/162 games.

4.5 OPS

In 1st place is OPS. This metric simply adds the last two stats together, OBP and SLG.

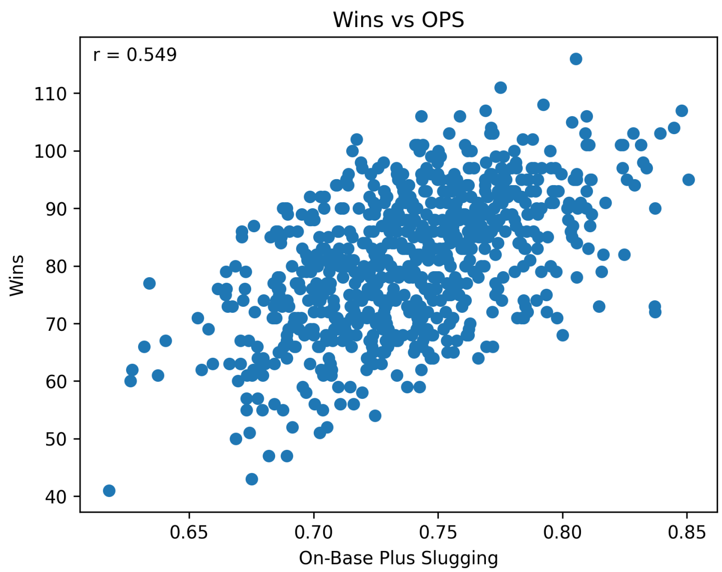

Figure 5: Relationship between OPS and regular season wins.

5 Discussion

The results of this analysis suggest that batting average is not nearly as strong of an indicator of team success as other modern metrics such as SLG, OBP and OPS. This supports the idea that simply recording hits is less informative that measuring the quality of those hits and a team's ability to generate total offensive production.

In particular, slugging percentage shows a noticeably stronger correlation with wins than batting average. This is because SLG accounts for extra-base hits, which contribute more directly to run scoring than singles. As a result, SLG better captures offensive power, which has a greater impact on a team's ability to win games.

On-base percentage performs even better than slugging in this analysis, highlighting the importance of avoiding outs. Since OBP includes hits, walks, and hit-by-pitches, it reflects a team's ability to sustain innings and create more scoring opportunities, which translates more directly into runs and wins.

OPS, which combines OBP and SLG, shows the strongest overall relationship with wins among the metrics tested. This suggests that a balanced measure of both getting on base and hitting for power provides a more complete representation of offensive value than any single component alone.

However, the differences between these correlations are relatively small and should not be overinterpreted. Wins in baseball are influenced by many factors outside of offense, including pitching quality, defense, bullpen performance, and situational sequencing. Therefore, even strong offensive metrics will only explain part of the variation in team success. This explains why the highest r-value observed was still less than 0.6 (not even an incredibly strong r-value itself).

Overall, the results suggest that while batting average remains a familiar and widely used statistic, it is less informative than more modern offensive metrics. Measures that incorporate power and on-base ability provide a clearer connection to run production and team success in the modern era of baseball.

6 Conclusion

This analysis shows that batting average is a relatively weak indicator of team success compared to more modern offensive metrics. While it remains a well-known and widely used statistic, it does not capture important aspects of offensive production such as power hitting or a team's ability to avoid making outs.

Metrics such as slugging percentage, on-base percentage, and especially OPS show stronger correlations with regular season wins, suggesting that overall offensive value is better represented by a combination of getting on base and hitting for extra bases rather than simply recording hits.

Overall, the results back up the trend in baseball analytics toward more detailed measures of player and team value by supporting the idea that modern era sabermetric statistics give a more accurate understanding of offensive performance than traditional batting average.